

High Cloud Airline Performance Analysis

Project Overview

This project analyzes airline flight operations and passenger performance using Microsoft Power BI.

The dashboard provides an interactive view of flight activity, passenger traffic, airline performance, load factor, routes, distance groups, and geographic patterns.

The objective is to convert raw airline data into meaningful business insights that can support operational and performance analysis.

Business Questions

- How does the airline load factor change over time?
- Which carriers have the highest passenger traffic?
- Which routes have the highest number of flights?
- How does flight activity vary by quarter and month?
- How does weekday performance compare with weekend performance?
- How does flight activity vary across distance groups?
- Which origin and destination locations contribute the most flight activity?

Dashboard KPIs

- Total Flights
- Total Passengers
- Available Seats
- Load Factor
- Total Carriers

Analysis Performed

Time-Based Analysis

- Monthly Load Factor
- Yearly Load Factor
- Quarterly Load Factor
- Weekday vs Weekend Load Factor

Airline Analysis

- Load Factor by Carrier
- Passenger Traffic by Carrier
- Number of Carriers

Route Analysis

- Top Routes by Number of Flights
- Flight activity by origin and destination
- Distance group analysis

Geographic Analysis

- Origin Country
- Origin State
- Origin City
- Destination Country
- Destination State
- Destination City

Tools & Technologies

- Microsoft Power BI
- Power Query
- Data Modeling
- DAX
- Interactive Slicers
- KPI Cards
- Charts & Visualizations

Dashboard Features

- Interactive filters
- KPI cards
- Monthly and yearly trend analysis
- Carrier comparison
- Route analysis
- Distance group analysis
- Geographic filtering
- Interactive dashboard navigation

Key Insights

The dashboard can be used to identify:

- Trends in airline load factor
- High-performing and low-performing carriers
- High-volume flight routes
- Passenger traffic patterns
- Differences between weekday and weekend operations
- Flight distribution across distance groups and locations

Project Structure

Airline-Performance-Analysis-PowerBI/

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|  
|— Airline_Performance_Analysis.pbix  
|— Dashboard-Executive.png  
|— Dashboard-Operations.png  
|— Dashboard-Analysis.png  
|— README.md
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Dashboard Preview

Executive Dashboard

Operations Dashboard

Analysis Dashboard

Skills Demonstrated

Power BI | Power Query | Data Modeling | DAX | Data Visualization | Business Analysis | Dashboard Development